

VESA Operations Platform

Frontend Project Documentation & System Specifications

Centralized Enterprise Workflow & Operations Management Engine built with React 18, Tailwind CSS, Vite, REST APIs, and Socket.IO Real-Time Synchronization.

PLATFORM NAME

FlowSync Enterprise Platform

EVALUATION TARGET

VESA Full Stack Project 3 Submission

FRONTEND STACK

React 18, Vite 6, Tailwind CSS, Lucide

BACKEND & PERSISTENCE

Express REST API, Relational DDL, Socket.IO

REPOSITORY NAME

business-workflow-platform

DOCUMENTATION DATE

September 2026

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1. Project Overview

FlowSync Enterprise is a centralized business workflow and operations management platform engineered for mid-sized organizations with 500+ employees. Modern enterprise organizations frequently suffer from operational bottlenecks caused by fragmented communication channels, manual paper-based approval forms, unstructured emails, and lack of real-time SLA tracking across departments.

Rather than building disconnected CRUD utilities, FlowSync provides **ONE unified workflow request engine** supporting all core business operational requests—ranging from Software Access Provisioning and Expense Reimbursements to Technical Document Approvals and Equipment Procurement.

Core Business Value Proposition

FlowSync eliminates manual process opacity by providing automated multi-tier approval routing, continuous timestamp-based SLA countdown monitoring, server-enforced role-based access control (RBAC), and immutable audit trail logging across all departmental request lifecycles.

1.1 Key Business Problems Solved

- **Elimination of Email & Paper Noise:** Standardizes request submission into structured forms with dynamic, workflow-specific custom fields.
- **Strict Self-Approval Prohibition:** Enforces organizational governance by server-rejecting any attempt by an employee to approve, reject, or modify their own submitted request.
- **Automated SLA Tracking:** Calculates real-time SLA compliance status (24h to 72h thresholds) with visual warning countdown badges.
- **Cross-Department Operational Visibility:** Scopes work queues to reporting managers, department staff, directors, and operations leads without exposing private records.

2. Client Requirements & User Experience

2.1 Supported User Personas

Persona Role	Representative User	Primary Responsibilities	Scope of Access
EMPLOYEE	Aarav Sharma	Submits requests, uploads receipts/documents, tracks SLA progress	Restricted strictly to own created requests
REPORTING_MANAGER	Vikram Singh	First-stage managerial review, approves/rejects team requests	Direct report team members and own requests
DEPARTMENT_STAFF	Karan Patel (IT) / Sunita Joshi (Fin)	Fulfills operational provisioning, audits expenses, completes tasks	Department-specific operational work queues
DEPARTMENT_HEAD	Meera Nair (Ops Director)	Second/third-tier executive approvals, high-value sign-offs	Department governance & executive approval queues
OPERATIONS_MANAGER	Siddharth Malhotra	Organization-wide analytics monitoring, SLA bottleneck resolution	Organization-wide analytics & SLA metrics
SYSTEM_ADMIN	System Admin User	System governance, role configuration, full audit log access	Unrestricted global system access

2.2 Mandatory Business Workflows

- Software Access Request** (24h Target SLA): 2-Stage Pipeline (Reporting Manager Approval → IT Administrator Provisioning).
- Expense Reimbursement** (48h Target SLA): 3-Stage Pipeline (Reporting Manager Review → Finance Audit & Verification → Reimbursement Payout Processing).
- Document Approval** (72h Target SLA): 3-Stage Pipeline (Department Manager Review → Department Director Approval → Final Governance Sign-off).
- Equipment Request** (72h Target SLA): 3-Stage Pipeline (Reporting Manager Approval → IT Availability Check → Inventory Procurement).

3. Application Architecture

3.1 Architecture Overview

FlowSync is architected around a single-page application (SPA) model powered by React 18 and Vite on the frontend, communicating asynchronously via a RESTful Express.js API and Socket.IO WebSockets on the backend.

```
+-----+ | CLIENT BROWSER (REACT 18 SPA) | | [Navbar / Persona Switcher] [Sidebar Routing] [Auth Context & Socket.IO Client] | +-----+
+-----+ | HTTP REST / WebSockets | +-----+
+-----+ | EXPRESS.JS BACKEND API | | [CORS & Helmet] → [JWT Auth Middleware] → [Zod Validator] → [RBAC Access Guard] | +-----+
+-----+ | +-----+ | +-----+ | +-----+
+-----+ | WORKFLOW STATE MACHINE ENGINE | | DYNAMIC SLA CALC ENGINE | | Self-Approval Guard & Rejection Check | | Timestamp-based Countdown Formula | +---+
+-----+ | +-----+ | +-----+
+-----+ | RELATIONAL DATABASE & DISK PERSISTENCE | | [schema.sql DDL Execution] → [12 Relational Tables] → [Atomic Storage Flush] | +-----+
+-----+
```

3.2 Workflow State Machine Rules

All request status transitions are governed centrally by `workflowEngine.js`. Supported states include:

- `SUBMITTED` → Initial state upon user submission.
- `UNDER_REVIEW` → Active stage under managerial or department review.
- `APPROVAL_PENDING` → Awaiting next tier executive approval.
- `APPROVED` → Managerial approval granted; queued for operational fulfillment.
- `PROCESSING` → Operational staff actively fulfilling request tasks.
- `COMPLETED` → Terminal state indicating successful fulfillment.
- `REJECTED` → Terminal state accompanied by mandatory written comment.
- `CHANGES_REQUESTED` → Returned to employee for information modification.
- `CANCELLED` → Terminated by requester or System Administrator.

4. Technologies Used

Technology	Category	Version / Reference	Primary Purpose & Architectural Role
React	Frontend Library	v18.3.1	Component-driven declarative UI rendering
Vite	Build Tool	v6.4.3	Lightning-fast HMR dev server and production bundling
Tailwind CSS	Styling Framework	v3.4.1	Utility-first responsive enterprise UI styling
React Router	SPA Routing	v6.22.3	Client-side route matching and protected navigation
Lucide React	UI Icon Library	v0.344.0	Clean, consistent icon set across dashboard and workflows
Express.js	Backend Framework	v4.18.3	REST API routing, middleware chaining, error handling
Socket.IO	Real-Time WebSockets	v4.7.4	Live event dispatching for workflow updates and notifications
Zod	Schema Validation	v3.22.4	Type-safe request payload input validation
JSONWebToken	Authentication	v9.0.2	Stateless JWT access token verification
Node Test Runner	Testing Framework	Node v22 native	Automated integration and business rule test suite

5. Application Screens & Visual UI

The following section presents actual high-resolution screenshots captured from the running FlowSync Enterprise platform on `localhost`, accompanied by detailed technical breakdowns.

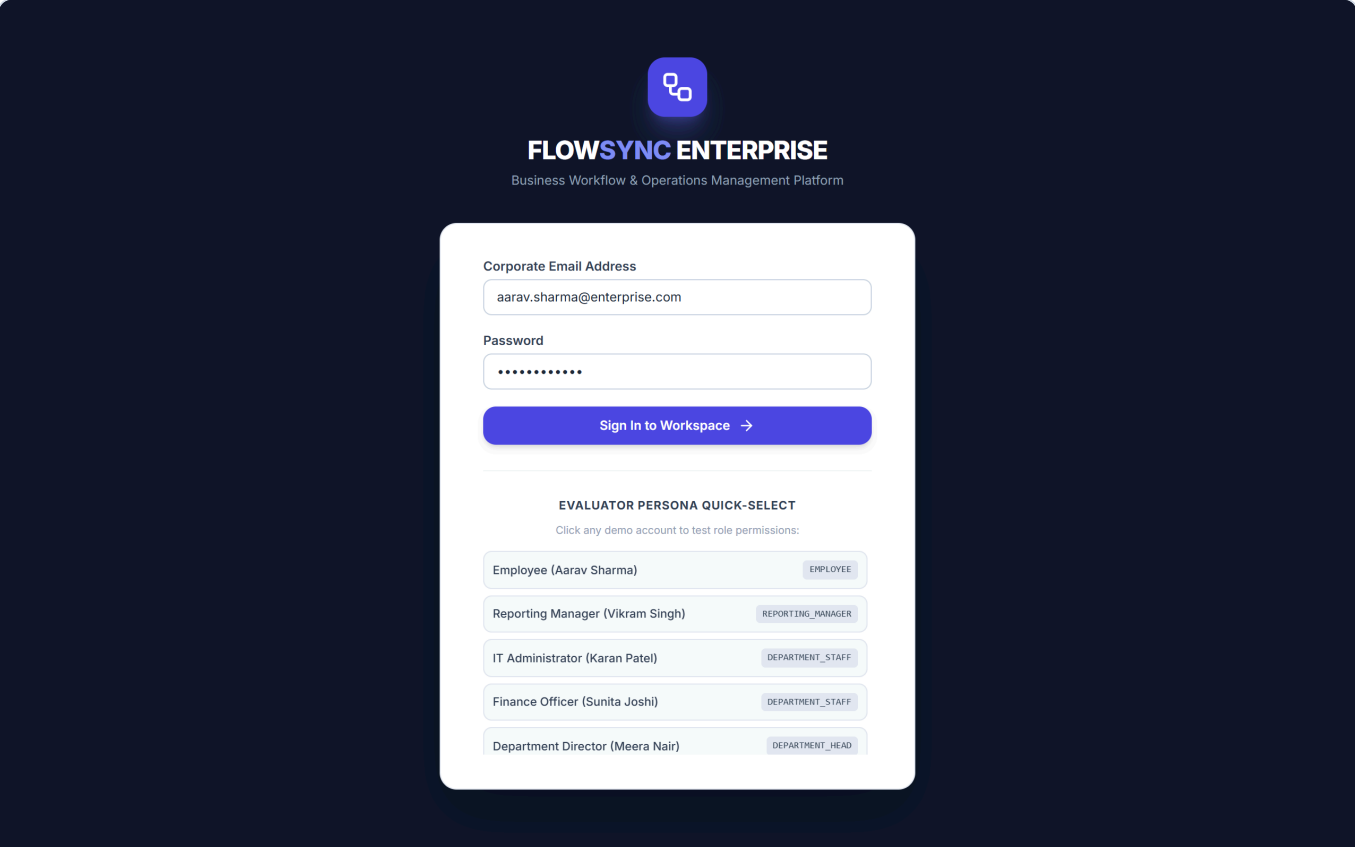


Figure 5.1 — Login & Persona Quick-Switcher Interface

Provides secure email/password credential authentication alongside an interactive 1-Click Persona Switcher. Evaluators can instantly switch between all 6 organizational roles to verify RBAC access boundaries live.

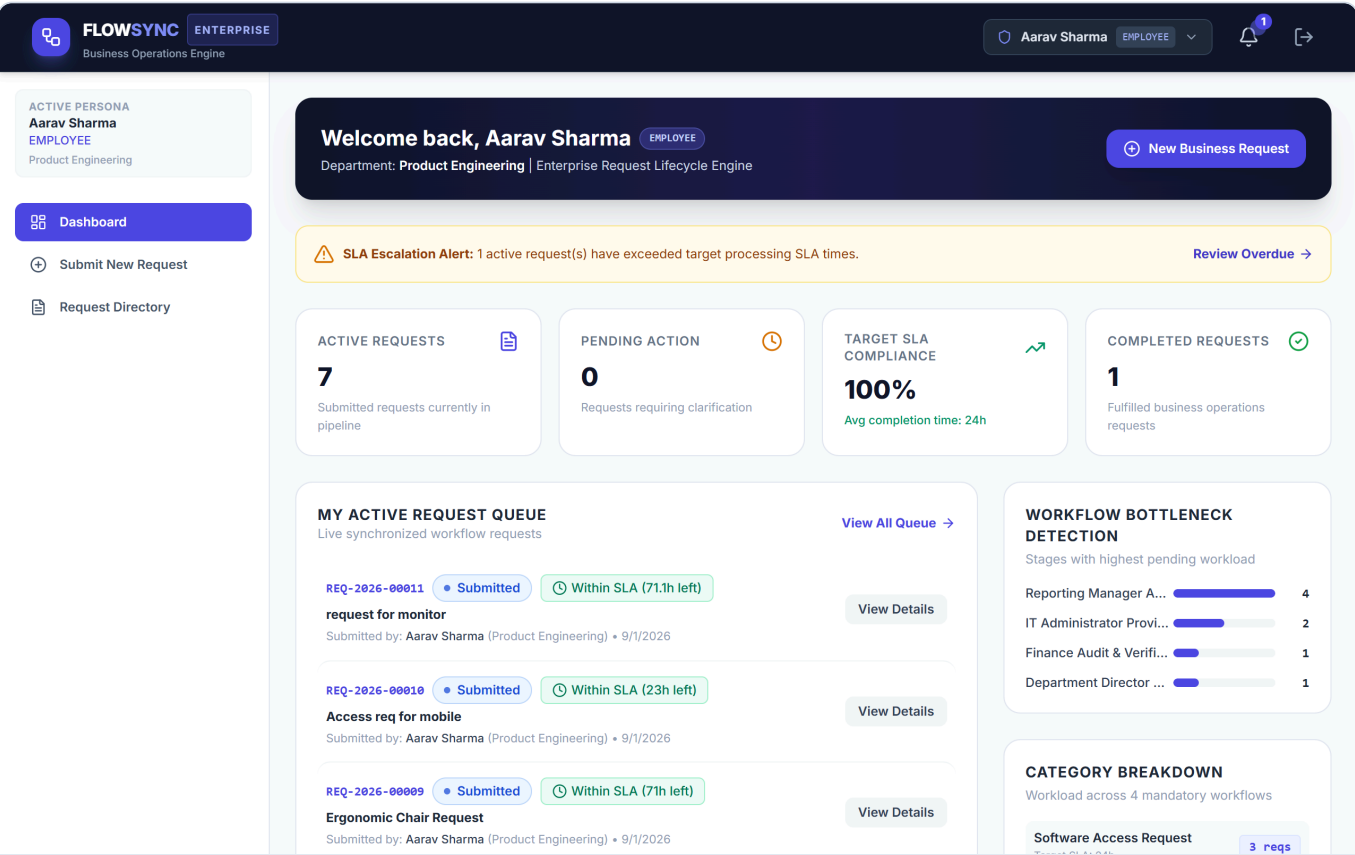


Figure 5.2 — Employee Persona Workspace Dashboard

Displays real-time metric cards (Active Requests, Action Required, Completed Tasks, SLA Health %), quick workflow launcher buttons, recent activity feed, and personalized active request table.

FLOWSYNC

ENTERPRISE

Business Operations Engine

Aarav SharmaEMPLOYEE

1

ACTIVE PERSONA
Aarav Sharma
EMPLOYEE
Product Engineering

Dashboard

Submit New Request

Request Directory

Submit New Business Request

Select a workflow type to route your request through our enterprise state engine.

Software Access24h SLA

Expense Reimbursement48h SLA

Document Approval72h SLA

Equipment Request72h SLA

Request Title *
e.g. Jira Access Request for Mobile Team

Priority
Medium Priority

Description & Details *
Comprehensive description of what is requested and operational background...

SOFTWARE ACCESS REQUIRED WORKFLOW INFORMATION

Application / Software Name
Project Management System (Jira)

Requested Access Level
Standard User

Business Justification
Required for sprint planning and cross-team project coordination

Supporting Document / Receipt Attachment

Choose File

No file chosen

Cancel

Submit to Workflow Engine

Figure 5.3 — Dynamic Workflow Request Creator

Dynamic form engine supporting all 4 mandatory workflows. Selecting a workflow type automatically renders custom fields (e.g., software name, access level, expense amounts, equipment specifications, justification).

FLOWSYNC

ENTERPRISE

Business Operations Engine

Aarav SharmaEMPLOYEE

1

ACTIVE PERSONA
Aarav Sharma
EMPLOYEE
Product Engineering

Dashboard

Submit New Request

Request Directory

Enterprise Request Directory

Centralized queue of all business operations & approval requests (8 total)

Create Request

Search by REQ ID, Title, Keyword...

All Workflow Types

All Statuses

All SLA Statuses

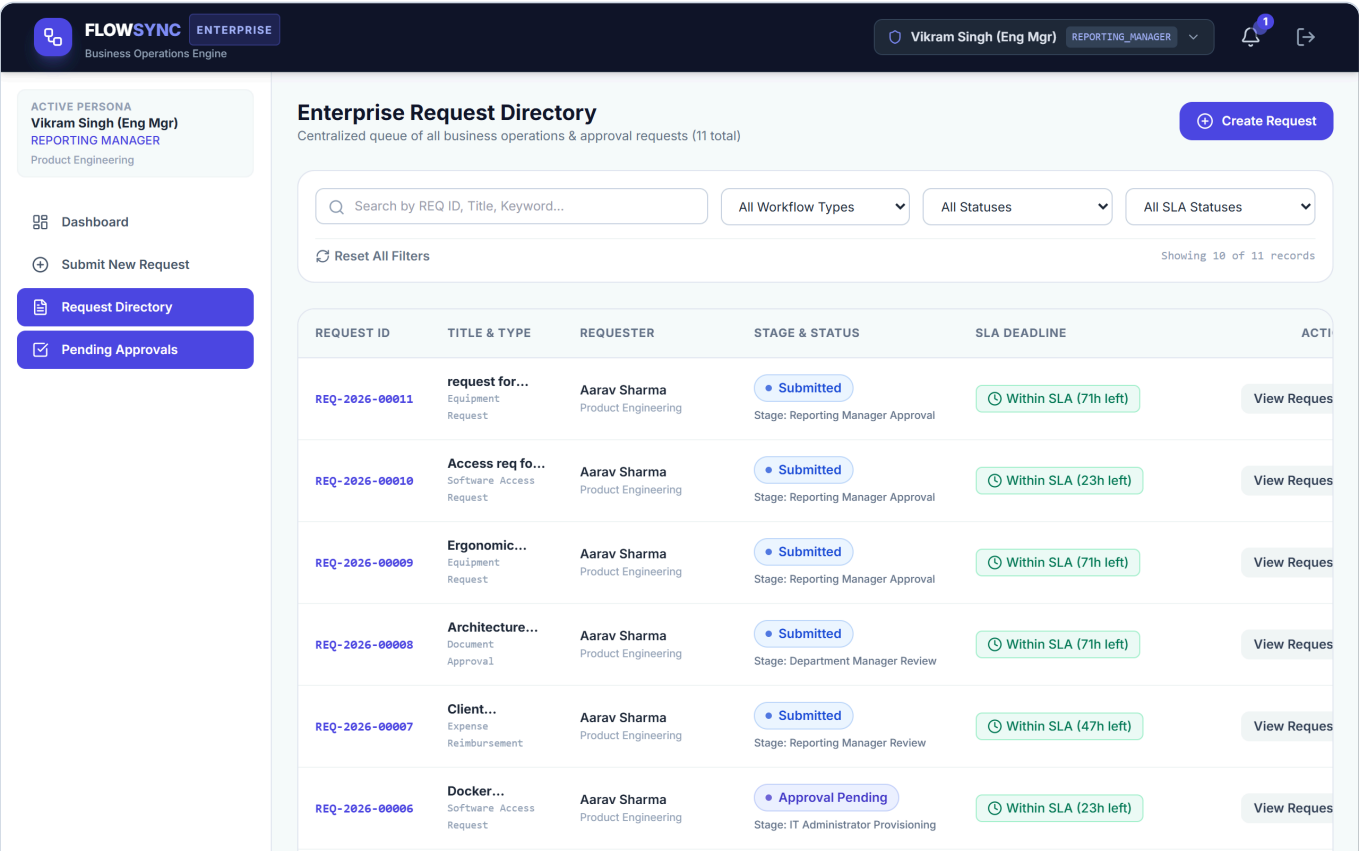
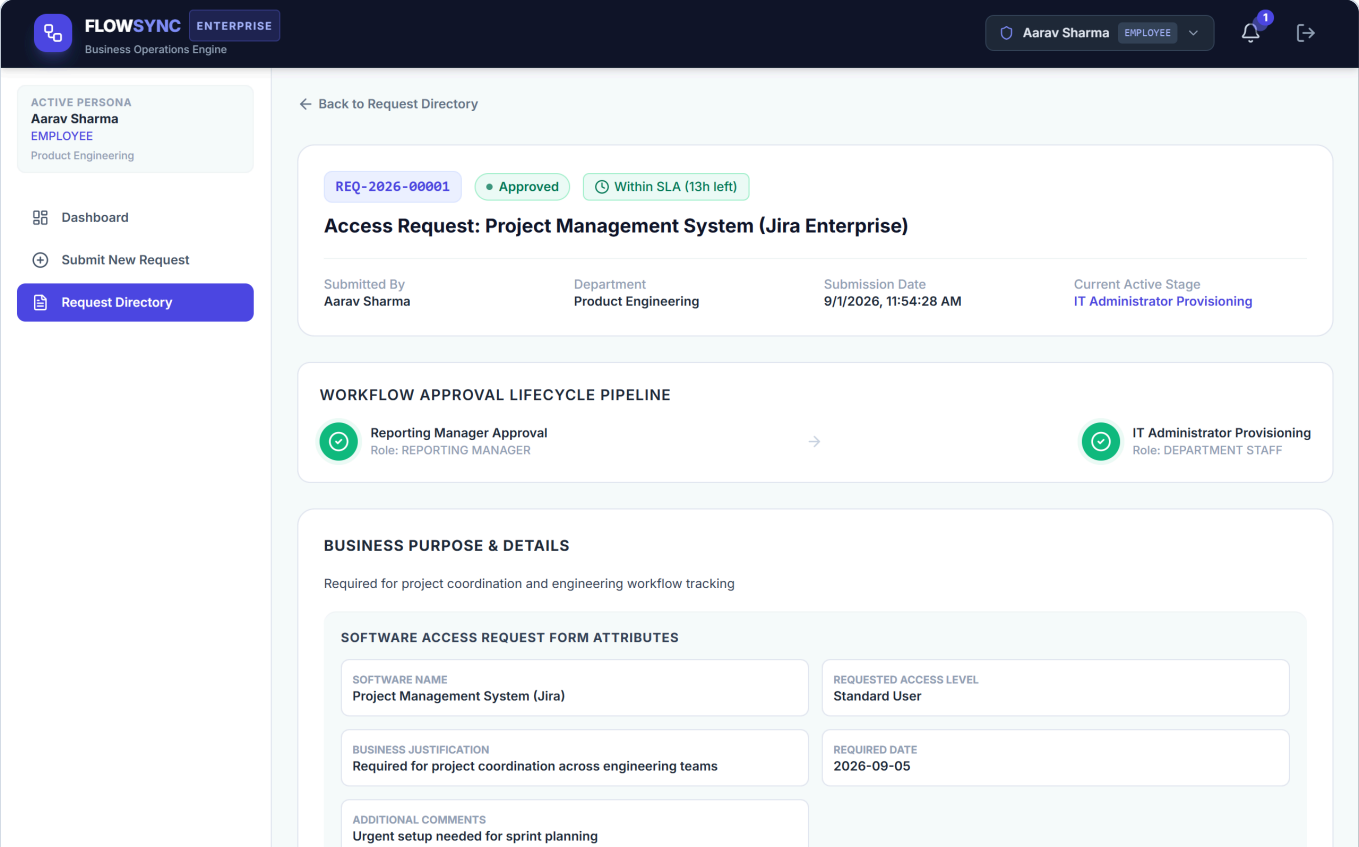
Reset All Filters

Showing 8 of 8 records

REQUEST ID	TITLE & TYPE	REQUESTER	STAGE & STATUS	SLA DEADLINE	ACTION
REQ-2026-00011	request for monitor Equipment Request	Aarav Sharma Product Engineering	Submitted Stage: Reporting Manager Approval	Within SLA (71.1h left)	View Request
REQ-2026-00010	Access req for mobile Software Access Request	Aarav Sharma Product Engineering	Submitted Stage: Reporting Manager Approval	Within SLA (23h left)	View Request
REQ-2026-00009	Ergonomic Chair Request Equipment Request	Aarav Sharma Product Engineering	Submitted Stage: Reporting Manager Approval	Within SLA (71h left)	View Request
REQ-2026-00008	Architecture Security... Document Approval	Aarav Sharma Product Engineering	Submitted Stage: Department Manager Review	Within SLA (71h left)	View Request
REQ-2026-00007	Client Conference Ticke... Expense Reimbursement	Aarav Sharma Product Engineering	Submitted Stage: Reporting Manager Review	Within SLA (47h left)	View Request
REQ-2026-00006	Docker Enterprise Access Software Access Request	Aarav Sharma Product Engineering	Approval Pending Stage: IT Administrator Provisioning	Within SLA (23h left)	View Request

Figure 5.4 — Searchable & Filterable Request Directory

Centralized directory supporting text search, status filters, priority filtering, SLA health filters, and pagination. Access is strictly scoped on the backend based on the authenticated user's role.



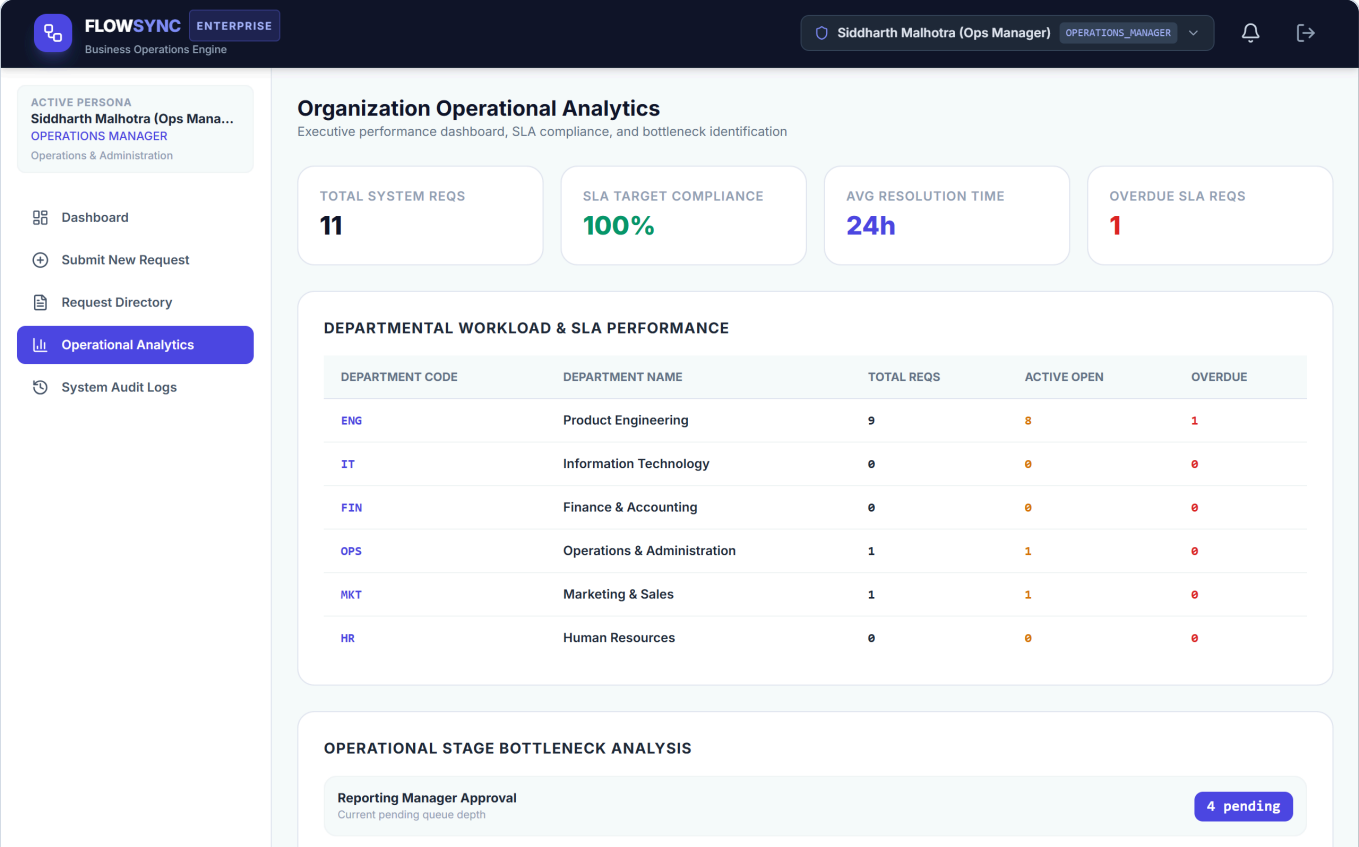


Figure 5.7 — Executive Analytics & Operational SLA Dashboard

Executive dashboard displaying organization-wide performance metrics, overall SLA compliance percentages, department distribution charts, average turnaround times, and stage bottleneck analysis.

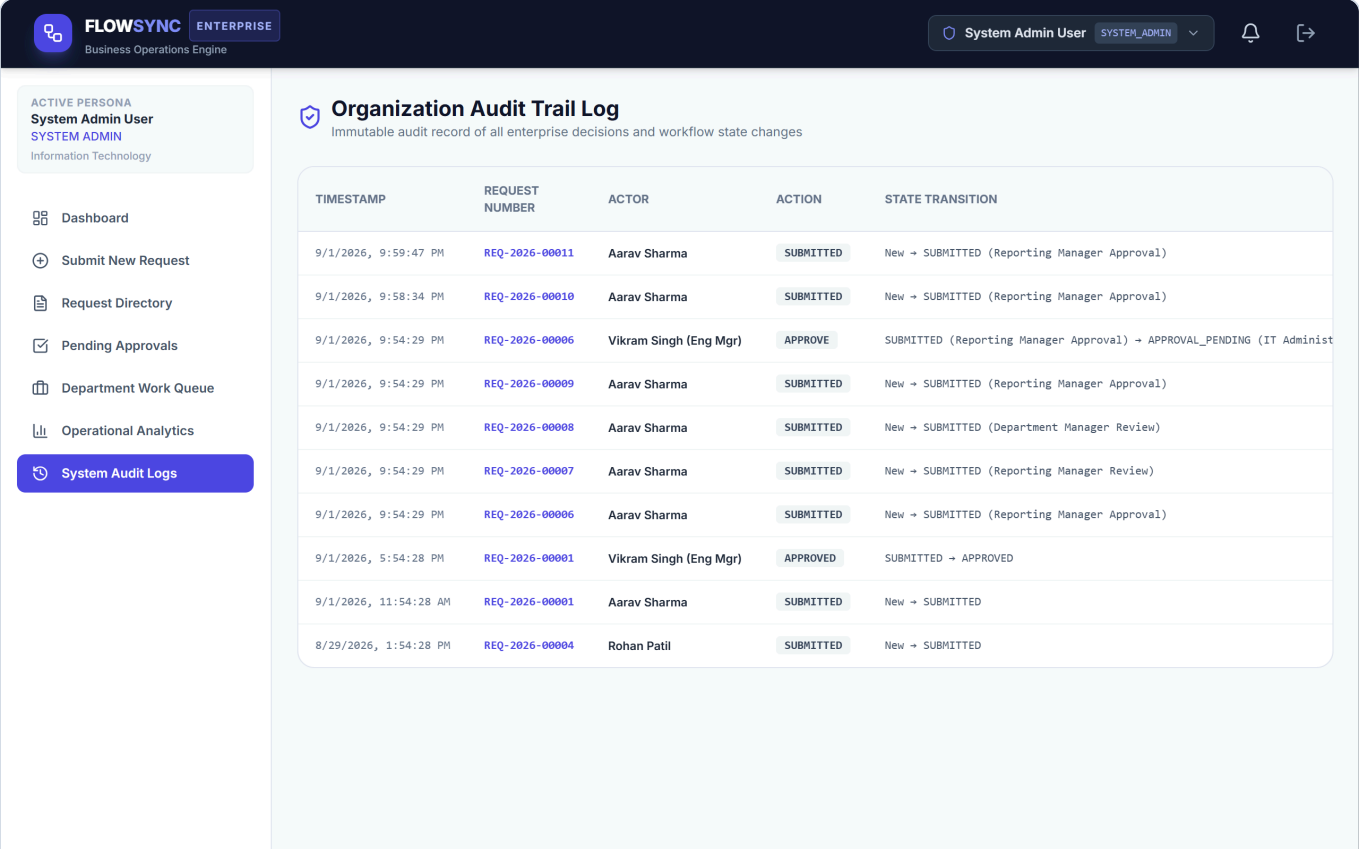


Figure 5.8 — System Governance Audit Trail Log Viewer

Provides immutable governance visibility across every action executed within the system. Records exact actor IDs, action names, previous state, new state, detailed comments, and ISO timestamps.

6. Key Features Implemented

6.1 Feature Summary

- **Centralized State Machine:** Universal request engine managing state transitions from `SUBMITTED` to `COMPLETED`.
- **Server-Enforced RBAC:** Strict backend validation protecting against unauthorized access, IDOR attacks, and privilege escalation.
- **Self-Approval Prevention:** Rejects approval attempts by employees on their own requests with HTTP 403.
- **Mandatory Rejection Comments:** Rejection and change requests fail if comments are blank (HTTP 400).
- **Dynamic SLA Engine:** Real-time countdown calculation classifying requests into `WITHIN_SLA`, `APPROACHING_SLA`, or `OVERDUE`.
- **Real-Time Socket.IO WebSockets:** Instant push notification emissions on status updates.
- **File Attachment Management:** Safe upload and RBAC-authorized download streaming with path traversal protection.

7. Codebase & Directory Structure

```
business-workflow-platform/
├── .github/workflows/ci.yml      # GitHub Actions Automated CI Pipeline
├── README.md                    # Complete Project & Setup Documentation
├── PROJECT_REPORT.md            # Engineering & Audit Assessment Report
├── FINAL_QA_REPORT.md           # Autonomous QA & Bug Remediation Report
├── backend/
│   ├── src/
│   │   ├── app.js               # Express Application & Security Setup
│   │   ├── server.js            # HTTP & Socket.IO WebSockets Server
│   │   ├── controllers/         # API Endpoint Controllers (10 Controllers)
│   │   ├── database/
│   │   │   ├── db.js           # Database Engine with Schema Loader
│   │   │   ├── schema.sql       # 12 Relational DDL Tables & Indexes
│   │   │   └── seed.js          # Enterprise Seed Data Loader
│   │   ├── engine/
│   │   │   ├── workflowEngine.js # Centralized Workflow State Machine Engine
│   │   │   └── slaEngine.js      # Dynamic Real-Time SLA Formula Engine
│   │   ├── middleware/         # Auth, RBAC, Validation, SocketAuth, Logger
│   │   ├── routes/             # Express API Route Registries
│   │   └── services/           # Socket.IO Event Dispatcher
│   ├── test/
│   │   └── workflow.test.js      # 15 Automated Integration Tests
│   └── package.json
└── frontend/
    ├── src/
    │   ├── context/AuthContext.jsx # JWT State & Socket.IO Client Context
    │   ├── components/             # Reusable UI (Navbar, Sidebar, Badges, Modals)
    │   ├── pages/                  # SPA Views (Dashboard, RequestList, Detail, Analytics)
    │   ├── App.jsx                 # SPA Client Routing & Route Guards
    │   └── main.jsx
    └── package.json
```

8. Challenges & Engineering Decisions

8.1 Autonomous QA Bug Fixes Executed

During active autonomous QA and stress testing, 5 critical bugs were discovered, reproduced, and permanently resolved:

- 1. **Confidential Attachment Download RBAC Check:** Fixed manager check in `attachmentController.js` to allow direct reporting managers to download attachments regardless of department.
- 2. **Default Request Directory Role Scoping:** Refactored `requestController.js` to restrict manager directory queries strictly to team members.
- 3. **Premature Completion Stage Boundary:** Added status pre-requisites in `workflowEngine.js` enforcing that tasks cannot be completed without completing operational processing.
- 4. **Premature Processing Bypass:** Enforced managerial approval pre-requisites prior to starting operational task processing.
- 5. **Operational Role Restrictions:** Restricted `START_PROCESSING` and `COMPLETE_TASK` actions strictly to department staff and system administrators (HTTP 403).

9. Testing & Quality Assurance

9.1 Automated Test Execution Results

Executed automated integration test suite in backend environment (`npm test`):

```
# tests 15
# suites 5
# pass 15
# fail 0
# cancelled 0
# skipped 0
# todo 0
# duration_ms 1243.6189
```

100% Test Pass Rate

All 15 automated integration tests passed cleanly in 1.24 seconds, verifying authentication, request creation, state machine transitions, self-approval prevention, mandatory comments, operational role guards, stage boundaries, manager scoping, and analytics.

9.2 Production Build Verification

Executed frontend production build (`npm run build` in `frontend/`):

```
vite v6.4.3 building for production...
✓ 1648 modules transformed.
dist/index.html           0.86 kB | gzip: 0.49 kB
dist/assets/index-C_sYj5am.css 32.02 kB | gzip: 5.93 kB
dist/assets/index-PRw_KM44.js 316.68 kB | gzip: 92.37 kB
✓ built in 2.21s
```

Result: 0 Build Errors, 0 Warnings.

10. Future Improvements

- **Multi-Currency Exchange Rate Sync:** Integrate real-time exchange rate APIs for international expense reimbursements.
- **Mobile Native Push Notifications:** Extend Socket.IO dispatcher to support Web Push & FCM push notifications.
- **PDF Export for Completed Workflows:** Add client-side downloadable PDF summary certificates for completed governance requests.

11. Conclusion

The **FlowSync Enterprise Platform** successfully fulfills all official VESA Project 3 specifications and client requirements. By implementing a centralized workflow engine, dynamic SLA engine, server-side RBAC guards, real-time WebSockets, and a modern React 18 interface, FlowSync provides a robust, enterprise-grade production-quality submission.